OXCARBAZEPINE

Hyponatraemia

Fewer interactions than carbamazepine

CHEMICAL NAME 10-oxo-10,11-dihydro-5H-dibenzo[b,f]azepine-5-carboxamide
CLASS Antiepileptic (keto-analogue of carbamazepine)
TRADE NAMES Trileptal · Oxtellar XR · Apydan

THERAPEUTIC USES / INDICATIONS

- Focal seizures with or without secondary generalisation (monotherapy or adjunct, adults and children ≥6 yr).
- Trigeminal neuralgia (off-label); bipolar disorder (off-label).

ADMINISTRATION

- **PO** twice daily with or without food; suspension shaken well.

DOSAGES

- **Adult:** start 300 mg twice daily; increase by up to 600 mg/day weekly; usual 600–2,400 mg/day.
- **Children ≥6 yr:** 8–10 mg/kg/day in 2 doses, up to 46 mg/kg/day.
- **CrCl <30:** start 300 mg daily.

PHARMACOKINETICS

- **Peak:** 1–3 h (MHD 4–12 h).
- Rapidly reduced to active monohydroxy derivative (MHD); no epoxide; minimal CYP induction (weak 3A4 inducer).
- Excreted in urine. **t½:** 2 h (MHD 9 h).

PHARMACODYNAMICS

- Blocks **voltage-gated sodium channels** → ↓ repetitive firing; modulates potassium and calcium channels.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dizziness, drowsiness, headache.
- Diplopia, ataxia.
- Nausea, vomiting.

ADVERSE EFFECTS (SERIOUS)

- Hyponatraemia (more frequent than carbamazepine).
- SJS/TEN (HLA-B*1502).
- Hypersensitivity (25–30% cross-reactivity with carbamazepine).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** hyponatraemia, diuretics, HLA-B*1502, AV block.

KEY INTERACTIONS

- **Hormonal contraceptives:** ↓ efficacy.
- **Phenytoin:** ↑ phenytoin levels.
- **Diuretics, SSRIs:** ↑ hyponatraemia.

NURSING MANAGEMENT

- Check Na^+ at baseline and during treatment (especially elderly).
- Ask about carbamazepine allergy.
- Monitor seizures, dizziness, rash.

PATIENT EDUCATION

- Report headache, confusion, drowsiness, nausea (low sodium).
- Use reliable non-hormonal contraception.
- Do not stop suddenly.

KEY POINTS

- Carbamazepine relative with fewer interactions.
- Hyponatraemia common.
- Focal seizures; 600–2,400 mg/day.
- Reduces pill efficacy.